Point-group symmetry

Point Group Tables

Point Group Tables of O_h(m-3m)

Click here to get more detailed information on the symmetry operations

Character Table of the group O_h(m-3m)*

O_h(m-3m)	#	1	4	2_{100}	3	2_{110}	-1	-4	m_{100}	-3	m_{110}	functions
Mult.	-	1	6	3	8	6	1	6	3	8	6	·
A_{1g}	Γ_1^+	1	1	1	1	1	1	1	1	1	1	$x^2+y^2+z^2$
A_{1u}	Γ_1^-	1	1	1	1	1	-1	-1	-1	-1	-1	·
A_{2g}	Γ_2^+	1	-1	1	1	-1	1	-1	1	1	-1	·
A_{2u}	Γ_2^-	1	-1	1	1	-1	-1	1	-1	-1	1	·
E_g	Γ_3^+	2	0	2	-1	0	2	0	2	-1	0	$(2z^2-x^2-y^2, x^2-y^2)$
E_u	Γ_3^-	2	0	2	-1	0	-2	0	-2	1	0	·
T_{2u}	Γ_5^-	3	-1	-1	0	1	-3	1	1	0	-1	·
T_{2g}	Γ_5^+	3	-1	-1	0	1	3	-1	-1	0	1	(xy,xz,yz)
T_{1u}	Γ_4^-	3	1	-1	0	-1	-3	-1	1	0	1	(x,y,z)
T_{1g}	Γ_4^+	3	1	-1	0	-1	3	1	-1	0	-1	(J_x,J_y,J_z)

The magnetic point group is m-3m', so it's pretty easy to see that the unitary (unprimed) subgroup is m-3. You can also check by going to MPOINT in Bilbao, put in m-3m', and get the listing of unprimed operators. These are summarized in the top row below, showing the operators in m-3:

Point Group Tables of T_h(m-3)

Click here to get more detailed information on the symmetry operations

Character Table of the group T_h(m-3)*

T_h(m-3)	#	1	3^+	3^-	2	-1	-3^+	-3^-	m	functions
Mult.	-	1	4	4	3	1	4	4	3	·
A	Γ_1^+	1	1	1	1	1	1	1	1	2 2 2

Returning to m-3m, we can classify operations at unitary/antiunitary, proper/improper:

		A			A	A		A	
					I	I	I	I	I
+	−	+	+	−	−	+	−	−	+

Antiunitary
Improper
Effect on theta

O_h(m-3m)	#	1	4	2_{100}	3	2_{110}	-1	-4	m_{100}	-3	m_{110}	functions
Mult.	-	1	6	3	8	6	1	6	3	8	6	·

So, theta transforms as a 1D irrep with these characters of +1 and -1. Looking at the chart at the top of the page, this is A_{2u}. So to couple at first order, we need an A_{2u} phonon.